Programming(4)

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Learning Objectives

- ✓ Understand how loops help repeat code efficiently
- ✓ Use for and while loops for repetition
- ✓ Apply break and continue to control loops
- ✓ Use an accumulator variable to keep running totals
- ✓ Compare for and while loops



Accumulators



Why do we need accumulators?

When we need to keep a running total (like sum, count, or average), we use a variable that “remembers” the previous value.

index.py > ...

```
1   total = 0
2   for i in range(5):
3       total = total + 1
4       print(total)
5
```

This variable total is called an accumulator

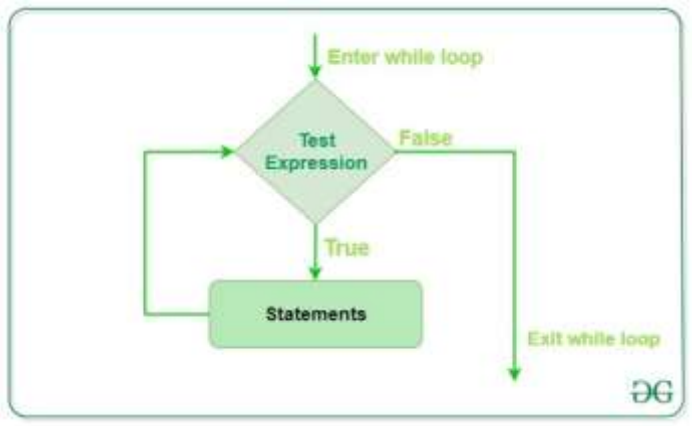


While



While

Repeats code *as long as a condition is True.*




```
3
4   x = 1
5   while x <= 5:
6       print(x)
7       x += 1
8
```



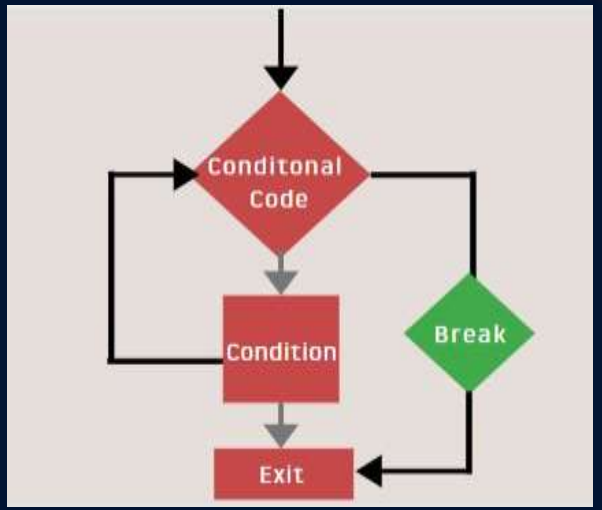
Break



Breaking out of a loop

break immediately stops the loop, no matter what the condition is.





For_example

index.py > ...

```python

for i in range(1, 10):
    if i == 5:
        break
    print(i)

```



Break

While_example

index.py > ...

```
1
2   while True:
3       name = input("Enter your name (stop to exit): ")
4       if name == "stop":
5           break
6       print("Hello,", name)
7
```



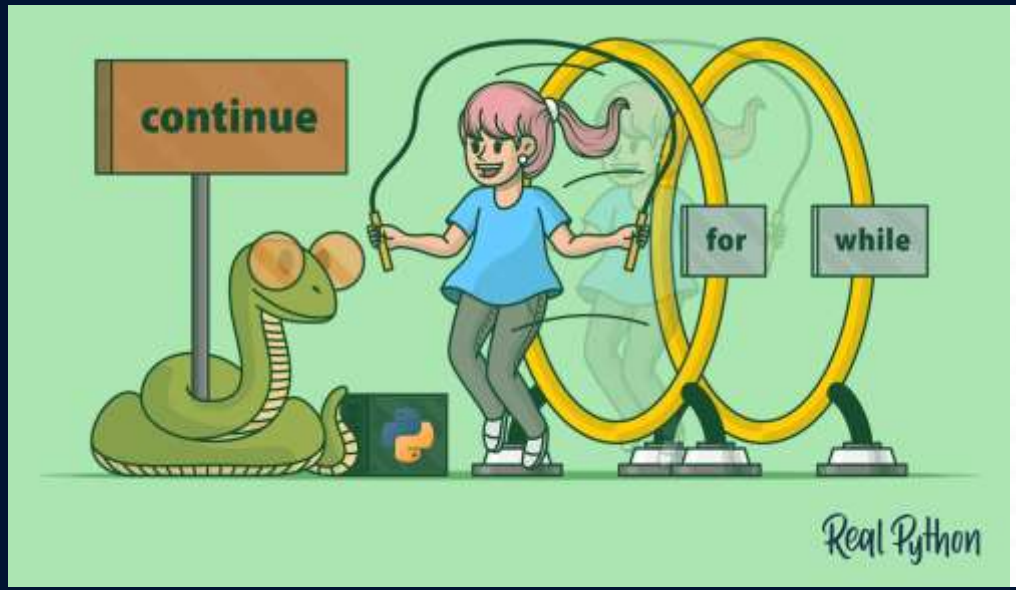
Continue

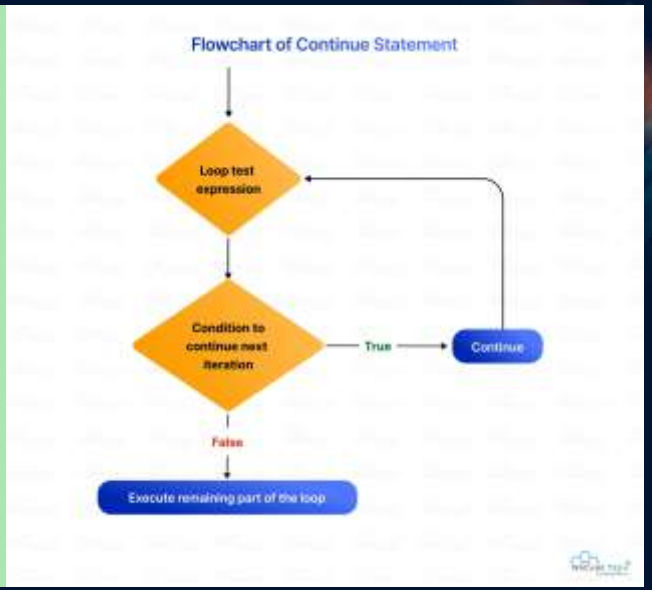
Continue

Skipping an iteration

continue jumps to the next loop cycle

(skips the rest of the code in this iteration).





Continue

For_example

index.py > ...

```
1
2   for i in range(1, 8):
3       if i == 4:
4           continue
5       print(i)
6
7
```



Continue

While_example

index.py > ...

```python

i = 0
while i < 10:
    i += 1
    if i % 2 == 0:
        continue
    print("Odd number:", i)


```



Thank You

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